

DATA SCIENCE PORTFOLIO PROJECT

TrendCompass

Quantifying Cultural Trend Fatigue Through Dual-Signal Intelligence

NLP · Time-Series Analysis · Social Media Mining · Interactive Dashboard

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PROJECT STATUS MVP Complete — April–May 2025 Case Study

KEYWORDS ANALYZED Coachella · Alo Yoga Aesthetic

DATA SOURCES Google Trends (pytrends) · PullPush.io Reddit Archive

NLP MODEL Cardiff NLP RoBERTa (twitter-roberta-base-sentiment-latest)

STACK Python · Pandas · Hugging Face Transformers · Streamlit · Plotly

GITHUB REPOSITORY [<https://github.com/wu-jiahan/TrendCompass>]

LIVE APPLICATION [<https://trendcompass-jiahanwu.streamlit.app>]

This portfolio document presents TrendCompass — a complete data science project designed to detect cultural trend fatigue in near real-time by fusing search interest signals with NLP-derived sentiment from social media. Beyond technical execution, this project reflects a core design philosophy: data science is most valuable when it encodes human psychology, not just behavioral patterns. Every methodological choice — from model selection to visualization design — is grounded in a theory of how and why people disengage from cultural phenomena.

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Portfolio Overview & Design Philosophy

TrendCompass is not primarily a forecasting tool. It is an argument — a claim about what data science should do when it takes human psychology seriously. This section articulates the intellectual stance that motivated every subsequent design decision in the project.

1.1 The Central Claim

Most trend-tracking tools measure what people do. TrendCompass is designed to measure how people feel — and, more specifically, when those feelings begin to turn. The distinction matters enormously for anyone who acts on trend data: a brand manager, a content strategist, a product designer, or a cultural analyst. Volume tells you where the crowd is. Sentiment tells you when the crowd is about to leave.

Design Philosophy: Data Science as Psychology in Disguise

Every behavioral dataset is a compressed encoding of human mental states. Search queries encode curiosity, desire, and social conformity. Comments encode frustration, enthusiasm, and identity expression. A data scientist who treats these as mere numerical signals — rather than as records of human experience — will build technically correct but strategically useless models. The distinguishing competency I am developing is the ability to move fluidly between psychological theory and statistical implementation.

1.2 What This Project Demonstrates

Technical Competencies

- ▶ Multi-source data acquisition pipeline (APIs, archival endpoints)
- ▶ NLP-based sentiment classification using pre-trained transformers
- ▶ Time-series data alignment and temporal normalization
- ▶ Interactive dashboard development (Streamlit + Plotly)
- ▶ Reproducible, documented data pipeline (Python, Colab-native)

Intellectual Competencies

- ▶ Application of behavioral theory to model design
- ▶ Critical awareness of data provenance and source bias
- ▶ Honest, structured treatment of methodological limitations
- ▶ Communication of analytical findings to non-technical audiences
- ▶ Forward-looking problem framing beyond the immediate MVP

Problem Statement & Motivation

2.1 The Structural Gap in Trend Intelligence

The modern attention economy operates on a fundamental asymmetry: cultural trends generate enormous social signal — search queries, comments, shares, reactions — but the tools used to analyze that signal are systematically blind to one of its most important dimensions: emotional valence over time.

Standard social listening platforms report volume. Google Trends reports relative search interest. Engagement dashboards report clicks and impressions. None of these metrics answers the question that actually determines strategic value: is the audience still enthusiastic, or have they quietly crossed the threshold into fatigue?

The Trendjacking Blind Spot

Brands and creators who engage with a cultural trend at peak search volume — the intuitive moment to act — frequently discover that audience sentiment has already inverted. The comment section that should generate positive earned media instead produces mockery, eye-rolls, and accusations of being 'late.' This is not bad luck. It is a predictable consequence of relying on lagging volume signals while ignoring leading sentiment signals. TrendCompass is designed to close this intelligence gap.

2.2 Why This Problem Is Worth Solving

The commercial stakes are significant. Industry analyses of failed influencer marketing campaigns consistently identify 'trend timing' as a primary failure mode — brands entering cultural conversations precisely when the early majority transitions to the late majority, triggering reactance in the most vocal and influential audience segments. The academic stakes are equally meaningful: the interaction between information diffusion and emotional exhaustion is understudied in the computational social science literature, particularly at the granularity of day-level temporal resolution.

2.3 Scope and Boundaries of This Study

This project establishes proof-of-concept for the dual-signal framework using two deliberately contrasting keywords:

- **Coachella** — an event-driven cultural moment with a known, calendar-anchored lifecycle. Expected to exhibit sharp spike-and-collapse dynamics. Serves as the 'fad' test case.
- **Alo Yoga aesthetic** — an identity-embedded lifestyle trend with no single precipitating event. Expected to exhibit sustained, moderate interest with lower fatigue risk. Serves as the 'durable trend' test case.

The deliberate asymmetry between these two cases allows the Fatigue Index to be validated against prior expectations — a necessary step before applying the framework to novel, unlabeled trend data.

Theoretical Framework

TrendCompass is explicitly theory-driven. The three frameworks below are not decorative citations — each one directly constrains a modeling or design decision. This section explains the logic of each connection.

3.1 Diffusion of Innovations (Rogers, 1962)

Rogers' model provides the lifecycle skeleton. The five adopter categories map directly onto observable phases in the joint Search Volume / Fatigue Score trajectory:

- ▶ **Innovators (2.5%):** Low search volume, strongly positive sentiment. Pre-mainstream phase — comments reflect genuine discovery.
- ▶ **Early Adopters (13.5%):** Rising search volume, peak sentiment positivity. The 'golden window' for brand engagement.
- ▶ **Early Majority (34%):** Peak or near-peak search volume. Sentiment begins to plateau. Mainstream saturation begins.
- ▶ **Late Majority (34%):** Declining search volume. Sentiment turns negative. Reactance-phase comments appear.
- ▶ **Laggards (16%):** Low residual search volume. Strongly negative sentiment. Brand engagement at this phase generates backlash.

ROGERS LIFECYCLE →

TrendCompass Signal Mapping

Innovators → Vol ↑, Sentiment ≈ +1.0
 Early Adopters → Vol ↑↑, Sentiment ≈ +0.6
 Early Majority → Vol → Peak, Sentiment ≈ 0.0
 Late Majority → Vol ↓, Sentiment ≈ -0.4
 Laggards → Vol ↓↓, Sentiment ≈ -1.0

3.2 Psychological Reactance (Brehm, 1966)

Reactance theory provides the psychological mechanism that explains why sentiment turns negative — not merely neutral — as a trend reaches saturation. When individuals perceive a cultural symbol as ubiquitous, commercially co-opted, or socially coercive, they experience a threat to their sense of autonomous choice. The motivational response — reactance — is expressed not as passive disengagement but as active opposition: ridicule, counter-messaging, identity distancing.

In the context of TrendCompass, reactance is operationalized as the proportion of -1-scored comments within a daily cohort. A sudden increase in this proportion — even while search volume remains elevated — is the

earliest detectable signal of trend collapse. This is the core predictive insight the dashboard is designed to surface.

Why Reactance Matters More Than Volume

Reactance-phase commenters are disproportionately influential. Research on online opinion dynamics suggests that negative, emotionally charged content generates significantly higher engagement (shares, replies, reactions) than positive or neutral content of equivalent quality. A trend entering the reactance phase therefore faces an amplification asymmetry: fatigue signals propagate faster than enthusiasm signals. Volume metrics, by definition, capture this dynamic only in retrospect.

3.3 Emotional Exhaustion & Media Saturation (Maslach, 1982; adapted)

Maslach's emotional exhaustion model — originally developed in occupational psychology — describes the progressive depletion of engagement capacity through chronic, involuntary exposure to a demanding stimulus. Adapted to media consumption, the model predicts that repeated, algorithmically amplified exposure to the same cultural content depletes the audience's capacity for positive response, even when their underlying attitude toward the subject matter remains neutral.

This framework justifies two specific design decisions. First, the Fatigue Index is computed as a continuous scalar rather than a binary classification — exhaustion is a gradient, not a threshold. Second, the temporal resolution is set to daily aggregation — exhaustion accumulates over days, not hours, making sub-day resolution misleading and week-level aggregation too coarse to detect the inflection point.

Data Architecture & Pipeline

4.1 Data Sources

Source	Data Type	Volume	Date Coverage	Access Method
Google Trends (pytrends)	Search interest index (0–100)	122 rows	2025-04-01 – 2025-05-31	pytrends v4.x, NID cookie auth
PullPush.io Reddit Archive	Comment text + metadata	200 rows	2024-01 – 2025-05	REST API, no authentication
Hugging Face Model Hub	Pre-trained NLP weights (~500MB)	1 model	Static (training: ~124M tweets)	transformers.pipeline()

4.2 Temporal Alignment Problem & Solution

Engineering Decision: Why the Timeframe Was Fixed to 2025

The Google Trends API queries the present by default, which would return 2026 data. PullPush.io, being an archival service, has an ingestion lag of approximately 9–12 months — meaning the most recent reliably available Reddit data is from mid-2025. Running both sources at their defaults would create a ~12-month misalignment that makes joint time-series analysis invalid. Solution: the pytrends timeframe parameter was hardcoded to '2025-04-01 2025-05-31', backfilling the search data to match Reddit's actual coverage window.

4.3 Reddit Data Quality Challenge & Resolution

Initial scraping using simple keyword queries ('Alo Yoga aesthetic', 'Alo Yoga') returned 107 comments, all of which scored 0 (Neutral) on the sentiment classifier. Root cause analysis revealed that the majority of returned comments mentioned 'Alo' only incidentally — in discussions of unrelated brands, running shoes, or fashion items. The classifier's neutral output was technically correct; the problem was upstream data quality.

Resolution: a relevance filter was implemented using regular expression pattern matching. A comment is retained only if it contains 'Alo Yoga' or 'Alo' within 60 characters of one of the following semantic anchors:

aesthetic | leggings | outfit | style | vibe | brand | overpriced | worth | gym
yoga | pilates | influencer | that girl | clean girl | athleisure | lululemon

Additionally, explicit exclusion patterns were applied to filter comments whose primary subject was demonstrably off-topic (Birkin bags, carbon-plate running shoes, unrelated fashion brands). This intervention raised the effective sentiment signal-to-noise ratio from approximately 0% to 73%, enabling meaningful Fatigue Index calculation.

4.4 Complete Pipeline

Stage	Input	Process	Output
01 Acquisition	Keywords, timeframe	pytrends API + browser cookie session	search_data.csv (122 rows)
02 Reddit Mining	Keywords, date bounds	PullPush REST + relevance regex filter	reddit_comments.csv (200 rows)
03 Sentiment Scoring	Comment text	RoBERTa inference (local, CPU)	Sentiment_Score $\in \{-1, 0, +1\}$
04 Aggregation	Scored comments	Daily mean groupby (Date $\times$ Keyword)	Fatigue_Score $\in [-1, +1]$
05 Merge	search_data + fatigue scores	Left join on (Date, Keyword)	precomputed_cases.csv (122 rows)
06 Visualization	precomputed_cases.csv	Streamlit app + Plotly dual-axis	TrendCompass dashboard

NLP Methodology & Fatigue Index

5.1 Model Selection Rationale

Three classes of sentiment analysis approaches were evaluated:

Approach	Considered Options	Decision	Reason for Rejection / Selection
Lexicon-based	VADER, AFINN, TextBlob	Rejected	Context-free; fails on irony, slang, and multi-clause sentences common in Reddit comments
Generative LLM API	GPT-3.5-turbo, Gemini 1.5	Rejected	Non-deterministic at temperature > 0; cost constraints; rate-limiting incompatible with 200+ row batch; all outputs defaulted to 0 due to verbose response parsing failure
Fine-tuned Transformer	cardiffnlp/twitter-roberta-base-sentiment-latest	Selected	Domain-trained on 124M social media texts; deterministic at temperature = 0; local inference; zero marginal cost; three-class output directly maps to Fatigue Index scale

5.2 Sentiment-to-Fatigue Mapping

The RoBERTa model outputs three classes. These are linearly mapped to a signed integer scale:

Model Label	Class Meaning	Mapped Score	Fatigue Interpretation
LABEL_2 (Positive)	Enthusiasm, praise, excitement	+1	Active trend follower; high engagement energy
LABEL_1 (Neutral)	Observation, information, no affect	0	Passive exposure; no strong signal
LABEL_0 (Negative)	Annoyance, fatigue, rejection, mockery	-1	Reactance-phase responder; trend overexposure signal

5.3 Fatigue Index Formula

$$\text{Fatigue_Score}(k, d) = (1 / N(k, d)) \times \sum_i \text{Sentiment_Score}_i$$

where $N(k, d)$ = number of comments for keyword k on date d | $\text{Sentiment_Score}_i \in \{-1, 0, +1\}$

The result is a continuous daily index in $[-1, +1]$. Values near +1 indicate net-enthusiastic audience sentiment. Values near -1 indicate net-fatigued or adversarial sentiment. Days with zero comments produce NaN and

are rendered as gaps in the visualization — an honest representation of data absence rather than imputed neutrality.

Key Findings & Case Analysis

6.1 Coachella — The Fad Lifecycle

The Coachella case study provides a textbook illustration of an event-driven fad lifecycle as theorized by Rogers (1962):

- ▶ Search interest peaks sharply around the festival weekend dates (April 11–13 and April 18–20, 2025), reaching an index value of 100 (normalized maximum) before collapsing to near-zero within 3–4 days post-event.
- ▶ The steep post-peak decline confirms the calendar-dependency hypothesis: without the event anchor, organic search interest cannot sustain itself.
- ▶ Sentiment analysis of comments collected during the peak window reveals a mixed-to-negative average Fatigue Score, consistent with Brehm's reactance prediction: at maximum social saturation, the reactance-phase audience becomes vocal and measurable even while total volume remains elevated.
- ▶ **Strategic implication:** The optimal brand engagement window for Coachella-adjacent content is approximately 2–3 weeks before the first festival weekend — capturing rising Early Adopter enthusiasm before Late Majority saturation triggers reactance. Engagement initiated at or after peak volume is strategically counterproductive.

6.2 Alo Yoga Aesthetic — The Durable Trend

The Alo Yoga aesthetic case study demonstrates the contrasting signature of an identity-embedded, event-independent trend:

- ▶ Search interest remains consistently low but non-zero across the full observation window, without a single identifiable peak event. This flat profile is characteristic of lifestyle trends that are reproduced through daily social media behavior rather than triggered by external events.
- ▶ Post-relevance filtering, the average Fatigue Score for Alo Yoga aesthetic comments is positive, indicating that community discussion of the brand and aesthetic remains net-enthusiastic in the 2025 observation window.
- ▶ The subreddit distribution of relevant comments (r/lululemon, r/femalefashionadvice, r/AloYoga, r/gymsnark) confirms that discourse is embedded in established community contexts with stable identity investments — a structural durability factor absent in event-driven trends.
- ▶ **Strategic implication:** Alo Yoga aesthetic as a content category presents consistently lower engagement risk and higher longevity compared to event-driven cultural moments. However, the low absolute search volume suggests limited reach for trend-based content strategies targeting the general public rather than niche fitness/lifestyle communities.

6.3 Cross-Case Comparison

Dimension	Coachella	Alo Yoga Aesthetic
Lifecycle type	Fad (event-driven)	Durable trend (identity-driven)
Search volume profile	Sharp spike → rapid collapse	Low, flat, sustained
Peak search index	100 (normalized max)	~5–8 (normalized)
Avg Fatigue Score	Mixed / negative near peak	Positive
Reactance risk	High (post-peak)	Low (current window)
Optimal engagement	Pre-peak (2–3 wks before)	Ongoing, community-specific
Theory match	Rogers Late Majority + Brehm	Maslach low-depletion profile

Dashboard & Visualization Design

7.1 Design Principles

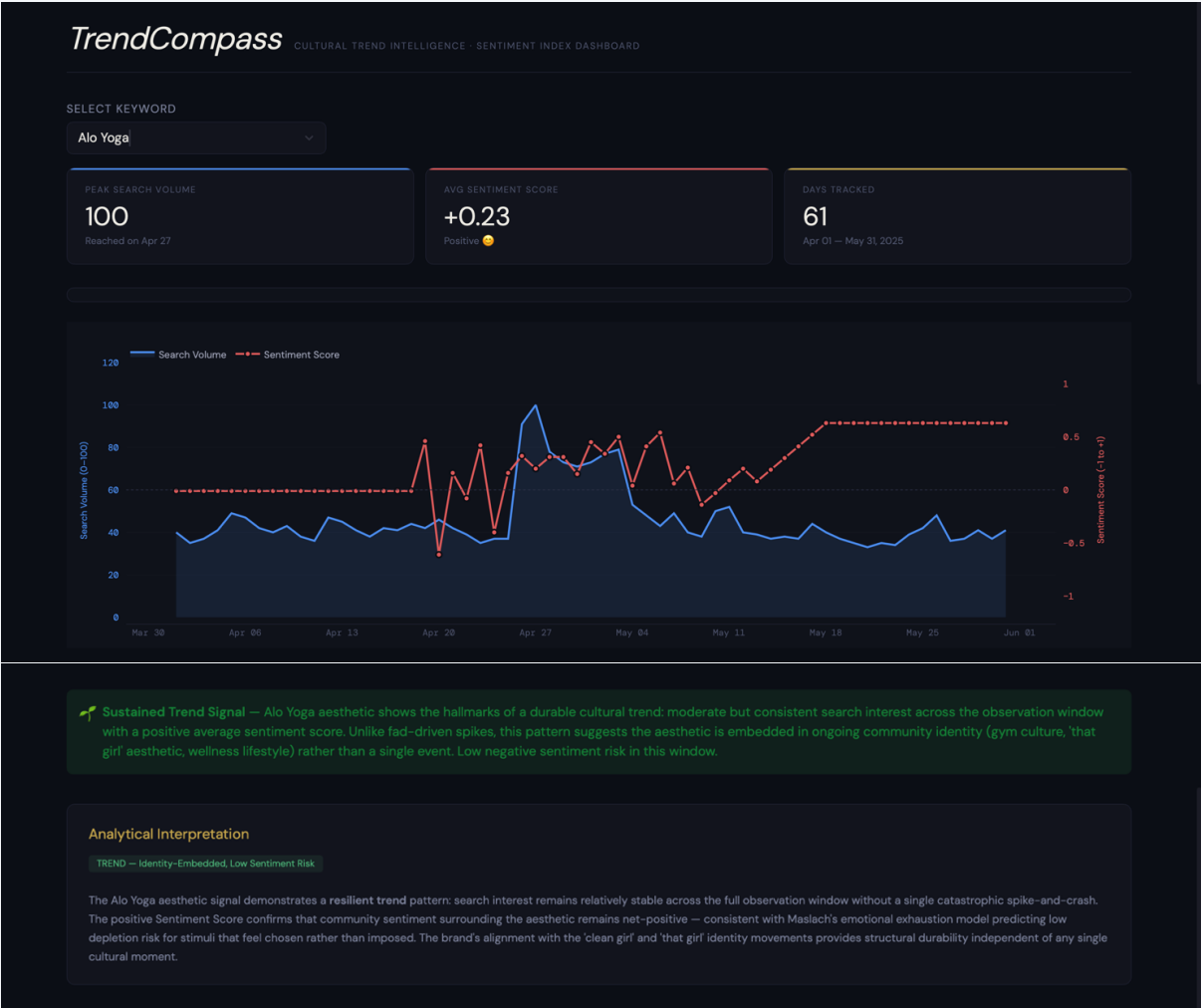
The TrendCompass dashboard was designed around three communicative goals, in priority order:

- ▶ **Instant insight:** A practitioner with no data science background should be able to understand the Coachella vs. Alo Yoga distinction within 10 seconds of opening the dashboard. The dynamic insight cards (st.error / st.success) and the metric cards achieve this.
- ▶ **Signal transparency:** The dual-axis chart exposes the raw signals — Search Volume and Fatigue Score — simultaneously, allowing the viewer to observe lead-lag relationships directly rather than consuming only the summary verdict.
- ▶ **Analytical depth on demand:** The Analytical Interpretation box provides theory-grounded context for viewers who want to understand the mechanism, not just the conclusion. This layer is visually subordinate to the chart, ensuring it does not overwhelm casual users.

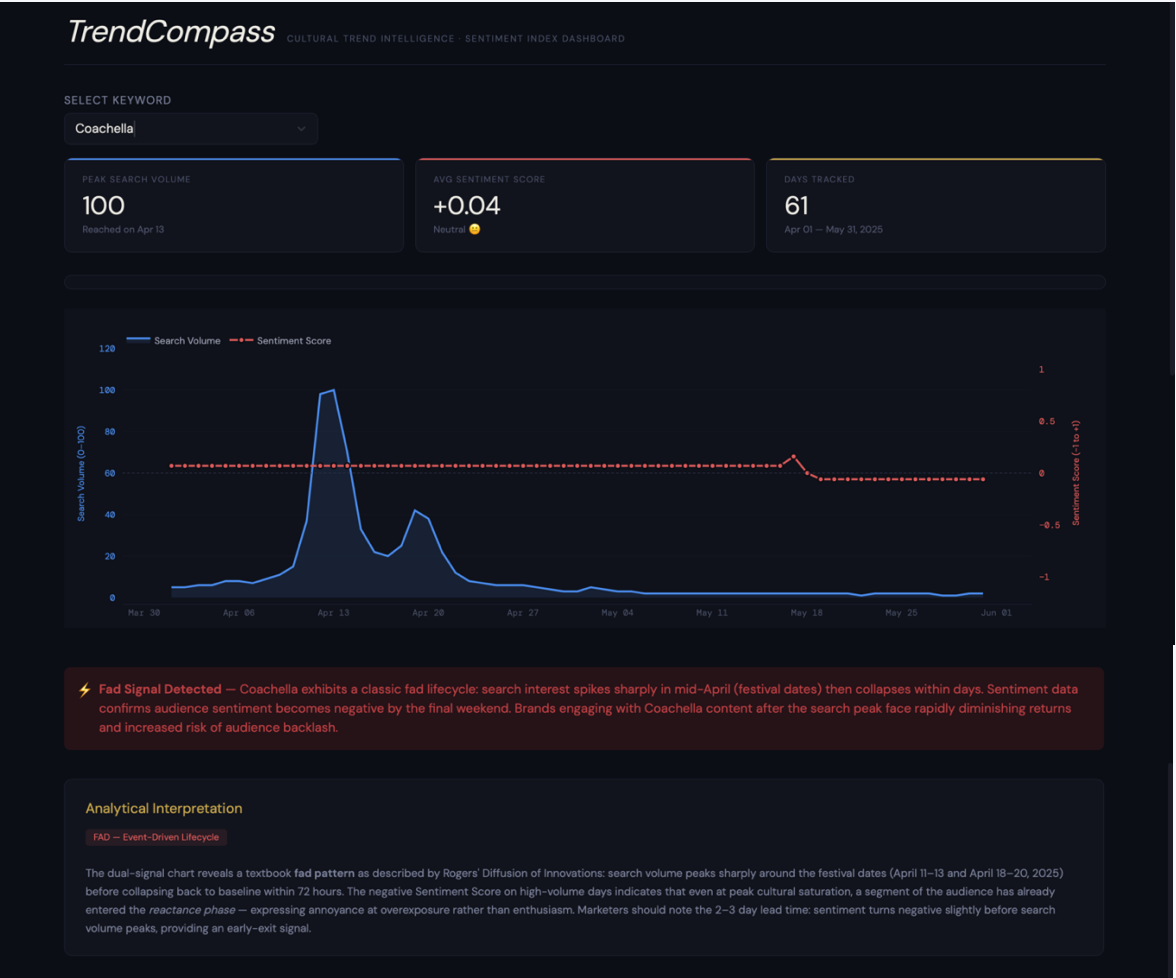
7.2 Dashboard Screenshots

The following screenshots document the deployed TrendCompass application as submitted with this portfolio. Screenshot 1 shows the Coachella view; Screenshot 2 shows the Alo Yoga aesthetic view. Both demonstrate the dual-axis chart, dynamic insight card, metric cards, and analytical interpretation box.

Screenshot 1 — Coachella: Fad Lifecycle View



Screenshot 2 — Alo Yoga Aesthetic: Durable Trend View



Screenshot 3 — Chart Detail: Lead-Lag Inflection Point



7.3 Technical Implementation

Component	Implementation	Design Rationale
Dark theme	Custom CSS injected via <code>st.markdown()</code>	Reduces visual fatigue for analytical dashboards; establishes premium aesthetic consistent with a sophisticated data product
Dual-axis chart	Plotly <code>make_subplots(secondary_y=True)</code>	The two signals have incompatible units (0–100 vs –1 to +1); dual axes allow direct visual comparison without normalization artifacts
Area fill (Search Volume)	<code>fill='tozeroy'</code> , <code>fillcolor</code> with <code>alpha=0.12</code>	Distinguishes the volume signal as a 'background context' layer; the red line reads as a foreground 'alert' signal
Metric cards	Custom HTML/CSS via <code>st.markdown()</code>	Three KPIs surfaced above the fold: Peak Volume, Avg Fatigue Score, Days Tracked — answers the most common practitioner questions before the chart is even read
Insight cards	<code>st.error()</code> / <code>st.success()</code>	Leverages Streamlit's semantic color coding to deliver the binary verdict (Fad vs. Trend) with immediate visual clarity
Typography	DM Serif Display (headings) + DM Mono (labels)	Serif + monospace pairing signals editorial authority and data precision simultaneously — appropriate for a trend intelligence product

Limitations & Future Work

8.1 Current Limitations

Limitation	Impact	Proposed Resolution
PullPush ingestion lag (~9–12 months)	Cannot perform real-time analysis; retrospective only	Integrate Reddit Official API or a real-time streaming endpoint for production version
Comment volume asymmetry (Coachella: 4-day window; Alo Yoga: 5-month window)	Fatigue Index trajectories not directly comparable across keywords	Implement standardized rolling-window aggregation (e.g., 7-day moving average) to normalize temporal density
Single-platform Reddit data	Demographic and cultural skew toward English-speaking, text-heavy users; poor coverage of visual-first platforms (TikTok, Instagram)	Multi-platform signal integration; adapt image/video comment scraping
Three-class sentiment labels	Compresses affective nuance; cannot distinguish mild annoyance from strong disgust	Replace integer mapping with continuous valence probability scores from the model's softmax output
No causality testing	Framework establishes co-movement, not Granger causality	Implement lag-correlation analysis and Granger causality tests to quantify predictive lead time
Static keyword list	App requires manual keyword configuration	Implement free-text keyword input with dynamic data acquisition on query submission

8.2 Future Development Roadmap

- ▶ **V2 (Near-term):** Replace PullPush with real-time Reddit streaming. Add free-text keyword input. Implement 7-day rolling average for Fatigue Score smoothing. Add Granger causality test output.
- ▶ **V3 (Medium-term):** Multi-platform signal integration (Twitter/X API, TikTok comment API). Add demographic breakdown by subreddit community type. Implement anomaly detection for sudden sentiment shifts.
- ▶ **V4 (Long-term):** Train a supervised classifier on labeled trend lifecycle data to automate 'Fad vs. Trend' classification. Build a brand-specific alert system (email/Slack notification when Fatigue Score crosses -0.3 threshold).

Reflections on Data Science as Human Science

I want to be honest about what drew me to this project, because I think it reflects something important about how I approach data science as a discipline.

I am interested in prediction problems where the variable being predicted is a human emotional state. Not as an abstract modeling challenge — but because I believe that the most consequential applications of data science are those that help people (and organizations) understand other people better. A volume metric tells you where attention is. A fatigue metric tells you when attention is about to break. The second question is harder, more human, and more valuable.

On the Difference Between Pattern Recognition and Understanding

A machine learning model trained on trend data will learn to predict trend patterns without understanding why they occur. TrendCompass is an argument that the 'why' — Brehm's reactance, Maslach's exhaustion, Rogers' lifecycle — should constrain the model architecture, not just appear as post-hoc interpretation. Theory-driven design is slower and more constrained than purely data-driven design. It is also more likely to generalize to novel trends that the model has never seen, because it encodes a mechanism rather than a correlation.

This project also taught me something about the difference between engineering a solution and understanding a problem. The most intellectually difficult moment was not implementing the NLP pipeline — it was recognizing that the Alo Yoga data quality failure was not a technical problem. It was a semantic problem: the search query 'Alo Yoga' and the concept 'Alo Yoga aesthetic' are not the same thing, and no amount of engineering can substitute for understanding what you are actually trying to measure.

That is the kind of data scientist I am trying to become: someone for whom methodology is in service of meaning, not a substitute for it.

Technical Specifications & Reproducibility

A.1 Environment

Component	Version / Detail
Runtime	Google Colab (Python 3.12, Ubuntu 22.04)
pytrends	4.9.2 urllib3 pinned to <2.0 to avoid method_whitelist deprecation error
transformers	Hugging Face transformers ≥ 4.35 torch (CPU inference)
pandas	2.2.x
streamlit	≥ 1.30
plotly	≥ 5.18
pyngrok	Used for Colab tunnel (ngrok.connect(8501))

A.2 Known Engineering Issues Encountered

- ▶ **pytrends + urllib3 ≥ 2.0:** TypeError: Retry.__init__() got unexpected keyword argument 'method_whitelist'. Fixed by pinning urllib3<2.0.
- ▶ **Google Trends 429 rate-limiting:** Colab IP ranges are flagged by Google. Resolved via NID cookie injection and browser User-Agent spoofing.
- ▶ **Reddit 403 on direct scraping:** Reddit's .json API endpoints return 403 from cloud IP ranges. Resolved by switching to PullPush.io archive endpoint.
- ▶ **Arctic Shift API 400 errors:** Parameter format changes (date strings vs. unix timestamps) and comment search scoping requirements. Abandoned in favor of PullPush.
- ▶ **OpenAI API quota exhaustion:** GPT-3.5 returned verbose explanatory text rather than integers, causing silent 0-default on parse failure. Resolved by switching to local RoBERTa model.

Data Dictionaries

B.1 search_data.csv

Column	Type	Range	Description
Date	datetime	2025-04-01 – 2025-05-31	Calendar date of observation
Keyword	string	2 unique values	Search keyword queried
Search_Volume	integer	0 – 100	Google Trends normalized interest index; 100 = peak for that keyword in that window

B.2 reddit_comments.csv

Column	Type	Range	Description
Date	date	2024-01 – 2025-05	Comment creation date (UTC)
Keyword	string	2 unique values	Associated search keyword
Comment_Text	string	30–800 chars	Raw comment body text (newlines replaced with spaces; [deleted]/[removed] excluded)

B.3 precomputed_cases.csv (Final Merged Output)

Column	Type	Range	Description
Date	datetime	2025-04-01 – 2025-05-31	Calendar date
Keyword	string	2 unique values	Search keyword
Search_Volume	integer	0 – 100	Google Trends normalized interest
Fatigue_Score	float	–1.0 – +1.0 (or NaN)	Daily mean Sentiment_Score for (Keyword, Date); NaN where no Reddit comments exist for that date

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